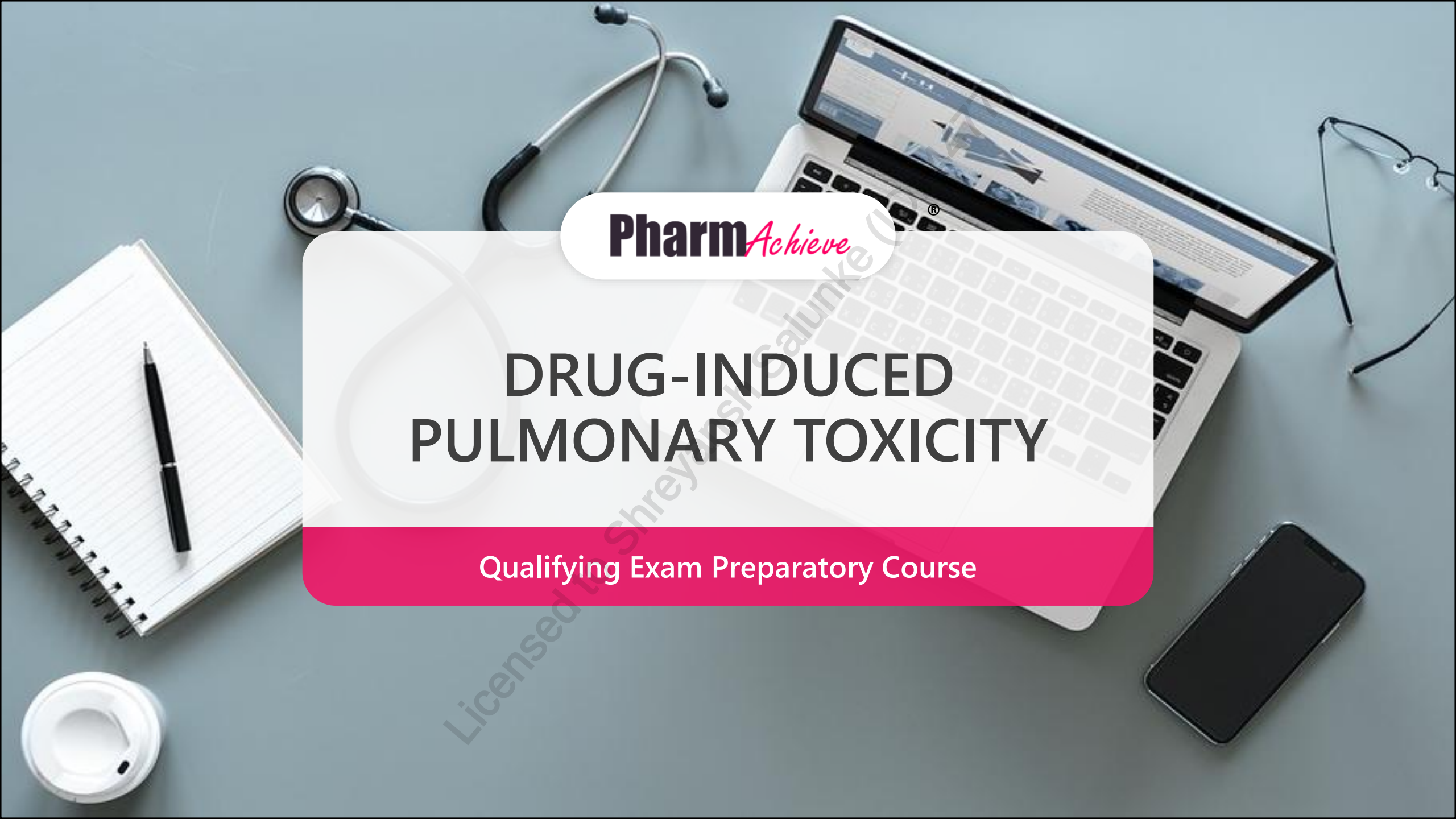


A top-down view of a grey desk with various items: a silver stethoscope, a laptop displaying a medical website, a pair of glasses, a black pen on a spiral notebook, and a white coffee cup.

Pharm*Achieve*

DRUG-INDUCED PULMONARY TOXICITY

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

ACE Angiotensin-Converting Enzyme

ALT Alanine Aminotransferase

AST Aspartate Aminotransferase

BPM Beats Per Minute

CBC Complete Blood Count

CRP C-Reactive Protein

DILD Drug-Induced Interstitial Lung Disease

ESR Erythrocyte Sedimentation Rate

HRCT High-Resolution Computed Tomography

IV Intravenous

NSAIDs Non-Steroidal Anti-Inflammatory Drugs

PCR Polymerase Chain Reaction

PO Per Os (Oral)

QHS Quaque Hora Somni (At Bedtime)

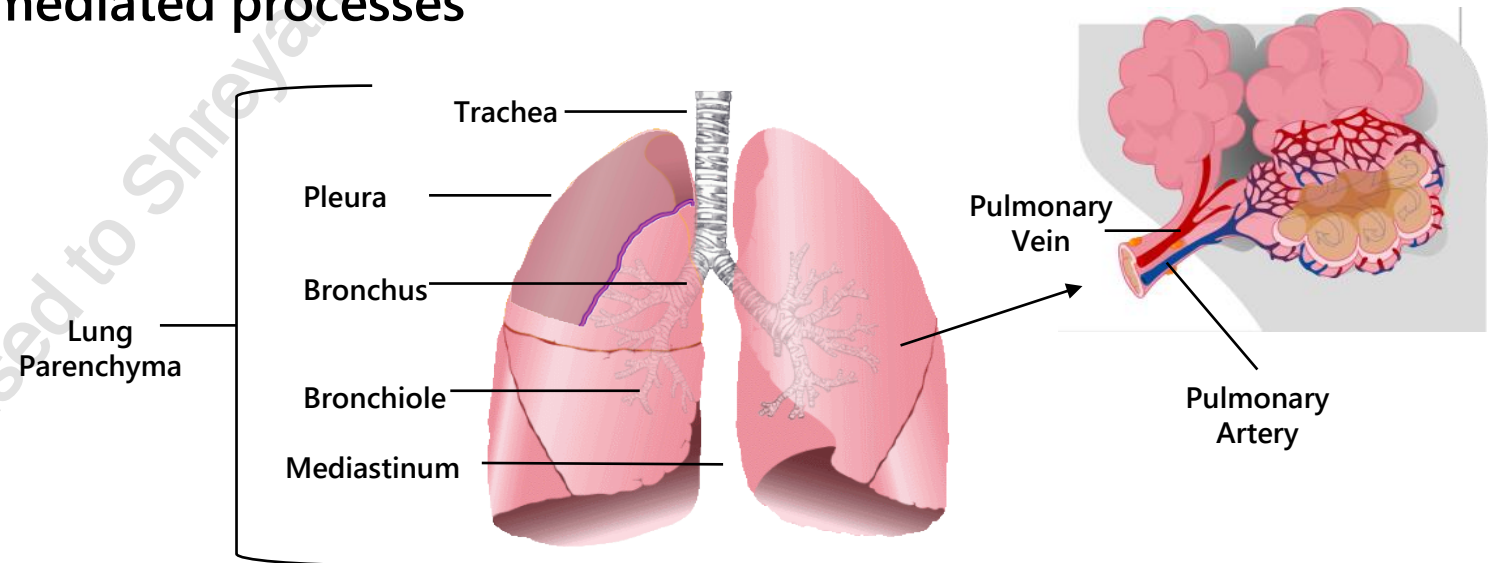
SpO₂ Oxygen Saturation

OBJECTIVES

- 1 Understand the clinical presentation of patients presenting with pulmonary toxicity
- 2 Identify common drugs that can cause pulmonary toxicity
- 3 Learn how to manage drug-induced pulmonary toxicity

INTRODUCTION AND ETIOLOGY

- Drug-induced pulmonary toxicity **can injure airways, pulmonary parenchyma, pleura, mediastinum and the pulmonary vascular system**
- Most **often causes drug-induced interstitial lung disease (DILD)**
- Proposed mechanisms include:
 - Increasing production of oxidants and inhibition of the anti-oxidant system
 - Cytotoxic damage from chemotherapeutic agents
 - Inflammatory and immune-mediated processes
 - Toxic drug metabolites



RISK FACTORS

- Children and elderly
- Combination therapy
- Cumulative dose
- Occupational factors
- Oxygen therapy
- Radiation
- Underlying lung disease
- Immune-mediated diseases (rheumatoid arthritis, systemic lupus erythematosus)
- Being immunocompromised
- Ethnicity/genetics
- Impaired renal/hepatic function

CLINICAL PRESENTATION

- Symptoms are non-specific
 - Cough, fever, dyspnea, wheezing, fever, pleuritic chest pain
- Diagnosis of exclusion
- Onset can be days to years after exposure
- Range of effects
 - Pleural effusion (accumulation of fluid in the pleural space)
 - Pneumothorax (collection of air in the pleural space between the lung and chest wall)
 - Pulmonary hemorrhage (bleeding from the pulmonary or bronchial vasculature)
 - Hemoptysis (the discharge of/"coughing up" blood from the lower respiratory tract)
 - Pulmonary vasculitis (inflammation and damage to blood vessels in the lungs)
 - Alveolar hemorrhage (bleeding into the alveolar spaces of the lungs)
 - Pneumonitis (inflammation of lung tissue) and fibrosis (formation of scar tissue in lungs)
 - Hypersensitivity
 - Bronchospasm (sudden constriction of muscles in the walls of the bronchioles)

PHARMACIST'S ROLE

Medication History

- Any history of similar symptoms?
- Changes in medication therapy?
- What are the doses of all the medications?
- Is the patient compliant?
- Have any medications been discontinued?
- Is there a possibility of overdose?
- When did the patient start taking his medications?

Exclusion of Other Primary or Secondary Lung Diseases

- Is there any medical condition that could be contributing to the symptoms?

Pulmonary Toxicity Symptoms

- After withdrawing the potential offending drug, does the patient go into remission?



Pharmacists need to obtain a thorough drug exposure history of the patient as there are **no diagnostic tests** available to confirm the presence of a drug-induced lung reaction

DRUG-RELATED CAUSES

Anti-Inflammatory Drugs

Aspirin® (acetylsalicylic acid), methotrexate, non-steroidal anti-inflammatory drugs (NSAIDs), sirolimus

Cardiovascular Drugs

Amiodarone, angiotensin-converting enzyme (ACE) inhibitors, beta-blockers, flecainide, hydralazine

Cytotoxic Drugs

Bleomycin, busulfan, carmustine, cyclophosphamide, methotrexate, mitomycin, vinca alkaloids, among others

DRUG-RELATED CAUSES

Antimicrobial Agents

Nitrofurantoin, sulfasalazine, sulfonamides, amphotericin B, isoniazid, ethambutol, tetracycline, minocycline

Controlled Substances

Cocaine, methadone, methylphenidate, heroin

Others

Biologics, carbamazepine, phenytoin, radiation, talc, tobacco

PHARMACOLOGIC TREATMENT

Step 1: Discontinue Offending Agent

- Acute reactions usually resolve within a day or two
- Chronic reactions may take longer and can be complicated by fibrosis that requires supportive care

Step 2: Manage Symptoms

- The goal is to stop the inflammatory response and prevent deposition of fibrotic tissue
- Patients may require respiratory inhalers or supplemental oxygen
- Corticosteroids may be useful for hypersensitivity pneumonitis

TAKEAWAY

Medication causes of drug-induced pulmonary toxicity include: anti-inflammatory drugs, cardiovascular drugs, cytotoxic drugs, antimicrobial agents, controlled substances, and biologics

Once the offending agent is discontinued, acute reactions resolve in 1-2 days

Chronic reactions may take longer to resolve and require supportive care

Symptoms may be managed with respiratory inhalers or supplemental oxygen. Corticosteroids may be useful for hypersensitivity pneumonitis

CASE STUDY

GS is a 64-year-old male who presents to the emergency department with complaints of progressive shortness of breath and a dry cough for the past 6 weeks. His past medical history is significant for hypertension, atrial fibrillation, benign prostatic hyperplasia, and dyslipidemia. He denies having any respiratory conditions. His current medications include ramipril 10 mg PO daily (initiated 3 years ago), amiodarone 200 mg PO daily (initiated 4 months ago), warfarin 6 mg PO daily (initiated 4 months ago), atorvastatin 80 mg PO daily (initiated 1 year ago), tamsulosin 0.4 mg PO QHS (initiated 2 years ago), and mirabegron 50 mg PO daily (initiated 2 years ago). GS has never smoked in his life, and he works in an office building as a chartered accountant. He reports that his BMI was 27.5 kg/m² 6 months ago, but 2 weeks ago he measured it to be 25 kg/m².

His vitals are recorded as: a respiratory rate of 22 breaths per minute, an oxygen saturation (SpO₂) of 95%, and a heart rate of 96 bpm. Laboratory test results returned this morning, revealing: an erythrocyte sedimentation rate (ESR) of 22 mm/hour (reference 0 to 15 mm/hr), a serum C-reactive protein (CRP) of 105 nmol/L (reference: ≤76.2 nmol/L), a serum alanine aminotransferase (ALT) level of 42 units/L (reference: 10 to 40 units/L), a serum aspartate aminotransferase (AST) level of 44 units/L (reference: 10 to 40 units/L), with normal complete blood count (CBC) values. Blood and sputum cultures are performed along with a viral polymerase chain reaction (PCR) test, indicating negative results for infections. Chest auscultation shows bilateral bibasilar crackles, a chest X-ray shows bilateral interstitial infiltrates, and a high-resolution computed tomography (HRCT) scan shows ground-glass opacities with diffuse interstitial thickening. After ruling out other possible causes, the physician suspects GS has drug-induced pulmonary toxicity.

CASE STUDY - QUESTIONS

1. Which of the following medications is most likely to be responsible for GS's symptoms?
 - a) Warfarin
 - b) Amiodarone
 - c) Atorvastatin
 - d) Ramipril
2. Which of the following therapeutic agents is the most appropriate to recommend?
 - a) Fluticasone/vilanterol inhaler
 - b) Oxygen inhaled
 - c) Montelukast PO
 - d) Methylprednisolone IV
3. Which of the following drugs is LEAST likely to cause drug-induced pulmonary toxicity?
 - a) Naproxen
 - b) Bleomycin
 - c) Amlodipine
 - d) Nitrofurantoin
4. Which of the following statements is true regarding drug-induced pulmonary toxicity?
 - a) It appears within 1-2 days of medication use
 - b) Its incidence is typically unrelated to a medication's dose or duration
 - c) It usually spontaneously resolves without intervention
 - d) Acute reactions may resolve in 1-2 days after the offending agent is discontinued

REFERENCES

1. Irely NS. Teaching monograph. Tissue reactions to drugs. *Am J Pathol*. Mar 1976;82:613-47.
2. Raissy HH, Harkins M. eChapter 15. Drug-Induced Pulmonary Diseases. In:DiPiro JT, Talbert RL, Yee GC, Matzke GR, Wells BG, Posey L, eds. *Pharmacotherapy: A Pathophysiologic Approach, 9e*. New York, NY: McGraw-Hill; 2014.
3. Wick J. Pulmonary Insult: Understanding Drug-Induced Lung Disease. Pharmacy Times. <https://www.pharmacytimes.com/publications/issue/2012/November2012/Pulmonary-Insult-Understanding-Drug-Induced-Lung-Disease>. Published 2012.
4. Cohen K, Salbu R, Addo-Atuah J, Rumore M. An Examination of Drug-Induced Pulmonary Disorders. Uspharmacist.com. <https://www.uspharmacist.com/article/an-examination-of-drug-induced-pulmonary-disorders>.
5. Chan ED, King TE. Amiodarone pulmonary toxicity. In: Flaherty KR, ed. UpToDate. 2024. <https://www.uptodate.com/contents/amiodarone-pulmonary-toxicity>.

CHANGE LOG

July 2025

- Formatting changes throughout
- Slide 3: Legend added
- Slide 6: Added 'impaired renal/hepatic function'
- Slide 9: Added 'acetylsalicylic acid' and NSAIDs, removed 'nitrosurea drugs' as carmustine already listed, added sirolimus to Anti-Inflammatory Drugs
- Slide 10: Added tobacco, removed sirolimus
- Slide 12: Takeaway added
- Slides 13, 14: Case study and questions added
- Slide 15: Updated references

October 2025

- No content changes made, minor formatting changes made

March 2026

- Slide 2: Updated NAPRA competencies
- Slide 9: Minor update